Ranking & route analysis

An explainable decision-support model; scientific goal and weights remain open.

PROPOSED • Not implemented • Scientific goal, scoring weights and hosting provider remain undecided

01

Required mission inputs

Jezero base, selected goal, walking crew profile, optional vehicle/robot support, time/range budget and return reserve. Include carried equipment, science work time and crew configuration. All capabilities are versioned assumptions awaiting agreement.

02

Candidate filtering

Rank reachable science stops within verified Jezero coverage, not distant crater centers. Stops may be outcrops, deposits or viewpoints. Verify coordinates before placement. Exclude stops outside round-trip limits or without required approach data.

03

Separate score components

Science value: evidence relevant to the chosen goal. Travel cost: route length, time and elevation change. Terrain difficulty: mapped slopes and obstacles. Data confidence: coverage and resolution; show it separately, not as a probability of safety.

04

Priority model — not yet defined

Normalize comparable features with fixed documented bounds. Version weights and feature definitions. Show each contribution and exclusion. Do not choose weights, thresholds or goals silently. Missing data stays unknown and must not improve a score.

05

Route engine — staged scope

Marswalk planning includes base-to-stops-to-base travel, stop work time, terrain concerns and assumptions. Use clearly labeled distance estimates first; bounded grid routing comes later. Field view shows numbered stops, science tasks and reference imagery.

06

Scientific validation

Check known examples, projection math, nodata behavior and budget exclusions. Test whether rankings change sensibly with goals and profiles. Evaluate sensitivity to weights and data resolution. No hazard probability, validated human route or certified safety rating is claimed.

Read technical-brief.md for linked sources, assumptions and competition timing.